	Question		Answer	Marks	AO	Guidance
3	(a)		Straight line drawn	B1 [1]	1.1	• Line through (3, 0) and (0, 6), mark intent • Condone good freehand line • Condone line stopping at axes • Ignore additional lines, perhaps as part of solution to (b) or (c), as long as it is not several attempts at (a)

Question			Answer	Marks	AO	• Guidance
3	(b)		A is $(3, 0)$ and B is $(0, 6)$	**B1**	**1.1**	• Coordinates soi by correct midpoint • Condone labels A and B reversed or omitted • Condone missing brackets for coordinates throughout if intent is clear
			Midpoint is $\left(\frac{3}{2}, 3\right)$	**M1**	**1.1**	• Correct midpoint or correct method to find the midpoint. e.g. $\left(\frac{3+0}{2}, \frac{0+6}{2}\right)$ • Allow embedded in the equation for perp. bisector
			Gradient of AB is -2	**M1**	**1.1**	• Correct gradient or correct method to find gradient e.g. $y = -2x + c$ or $\frac{6-0}{0-3}$ or $\frac{6-3}{0-1.5}$ or $\frac{0-3}{3-1.5}$ oe • FT *their* midpoint if used • Implied by correct gradient of perpendicular bisector
			Gradient of perpendicular bisector is $\frac{1}{2}$	**M1**	**1.1**	• FT $\frac{-1}{\textit{their}\ \text{gradient}}$ • May be seen embedded in the equation.
			Equation $(y - 3) = \frac{1}{2}\left(x - \frac{3}{2}\right)$	**M1**	**1.1**	• Using *their* midpoint and *their* gradient of perpendicular bisector • Allow $y = \frac{1}{2}x + c$ **and** $3 = \frac{1}{2} \times \frac{3}{2} + c$ oe
			$y = \frac{1}{2}x + \frac{9}{4}$	**A1**	**1.1**	• OR $2x - 4y = -9$ • Allow any correct equivalent simplified form e.g. $4y = 2x + 9$ or $y - \frac{1}{2}x = \frac{9}{4}$ or $\frac{1}{2}x - y + \frac{9}{4} = 0$ or any multiple of these. • Allow decimals e.g. $y = 0.5x + 2.25$ • isw after correct equation seen
				[6]		

Question			Answer	Marks	AO	Guidance
3	(c)		$(AB^2 = a^2 + a^2 =)3^2 + 6^2$ Or $(AC^2 =)1.5^2 + 3^2$ where C is the midpoint of the square	B1	3.1a	• Use of Pythagoras which will help to calculate side e.g. AC^2, BC^2, AB^2 or AC, BC, AB soi • FT *their* midpoint from (a) if used
			$a^2 + a^2 = 45$	M1	1.1	• Further working may not be seen as likely to be done by calculator
			Alternative method 1 for M1 $a^2 = 2AC^2 = 2(1.5^2 + 3^2)$	M1		• Using $AC = \sqrt{1.5^2 + 3^2}$ where C is the midpoint of the square
			Alternative method 2 for M1 $a = \sqrt{3^2 + 6^2} \times \cos 45°$ oe	M1		• Use of trigonometry to find *a*
			Alternative method 3 for M1 $a = \frac{\sqrt{1.5^2+3^2}}{\cos 45°}$ oe	M1		• Use of trigonometry to find *a* • Using $AC = \sqrt{1.5^2 + 3^2}$ where C is the midpoint of the square
			Alternative method 4 for M1 $(x - 1.5)^2 + (0.5x - 0.75)^2 = 1.5^2 + 3^2$ $4x^2 - 12x - 27 = 0$	M1		• Use of *their* AC and *their* equation of perpendicular bisector to form a quadratic equation to give the *x*-coordinates of the other two vertices of the square
			$\frac{3}{2}\sqrt{10}$ oe	A1	1.1	• Accept any exact correct answer e.g. $\frac{3\sqrt{10}}{2}$, $\frac{3}{4}\sqrt{40}$, $1.5\sqrt{10}$
			Alternative method using vectors $(\overrightarrow{CB} =) \begin{pmatrix} -1.5 \\ 3 \end{pmatrix}$ so $(\overrightarrow{CD} =) \begin{pmatrix} 3 \\ 1.5 \end{pmatrix}$ where D is another vertex of the square.	B1		• FT *their* midpoint from (a) • Vector from C to B or A or vice versa **and** rotating to get vector to one of the other two vertices
			D is $(4.5, 4.5)$ so $a^2 = (4.5^2 + 1.5^2)$	M1		• Or $(-1.5, 1.5)$
			$\frac{3}{2}\sqrt{10}$ oe	A1	1.1	• Accept any exact correct answer e.g. $\frac{3\sqrt{10}}{2}$, $\frac{3}{4}\sqrt{40}$, $1.5\sqrt{10}$
				[3]		